

National Institute of Technology, Rourkela

Name of the Examination: B. Tech. End-Semester (Spring 2021-22)

Branch : CS

Semester : VI

Title of the Course : Computer Networks

Course Code : CS3002

Time: 3 Hours

Maximum Marks: 50

- Instructions:** 1. Answer **all** short-answer type questions from **Section-A**.
2. Answer any **four (4)** questions from **Section-B**.
3. All parts of a question must be answered at one place.
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Section-A

(Attempt all short-answer type questions from this section)

1. Answer the following parts. (10 x 1 = 10 Marks)
- ~~(a)~~ What is the optimality principle used for routing algorithm?
 - (b) Answer True or False for the following statements
 - (i) DHCP always assigns a client an IP address for an *indefinite* period of time.
 - (ii) The logical address will change from hop to hop, but the physical addresses usually remain the same?
 - (c) A class B network on the Internet has subnet mask of 255.255.240.0. What is the maximum number of hosts per subnet?
 - ~~(d)~~ What is an IP *Prefix*?
 - (e) What is loose source routing?
 - (f) Give any one advantage and any one disadvantage of the Flooding?
 - (g) Why *Traffic Shaping* is required in the network?
 - (h) What is *multicasting*?
 - (i) Explain the situation when the routing protocol leads to count-to-infinity problem?
 - (j) Give two reasons why Network Address Translation is so important?

Section-B

(Attempt any four from this section)

(10 x 4 =40 Marks)

2. Answer the following.
- (a) Derive an equation for channel efficiency of classic Ethernet in terms of the frame length, **F**, network bandwidth, **B**, the cable length, **L**, and the speed of signal propagation, **c**, for the optimal case of **e** contention slots per frame under conditions of heavy and constant load, that is, with **k** stations always ready to transmit. Assume a constant retransmission probability in each slot. (5 Marks)
 - ~~(b)~~ Explain Switched Ethernet using a neat diagram. Explain why CSMA/CD algorithm is not needed in it? (5 Marks)

3. Answer the following:
- (a) Explain the following techniques: **(5 Marks)**
 - (i) Leaky bucket algorithm
 - (ii) Packet scheduling algorithm
 - (b) Explain Load Shedding approach used for network layer congestion control? **(5 Marks)**
4. Answer the following:
- (a) Explain the following two TCP congestion control algorithms: **(5 Marks)**
 - (i) Slow Start
 - (ii) Fast Retransmit and Fast Recovery
 - (b) Explain Internet Control Message Protocol. Describe the important ICMP message types? **(5 Marks)**
5. Answer the following:
- (a) Explain two-army problem. Explain the handshaking mechanism of TCP to release a connection between two hosts in a network? **(5 Marks)**
 - (b) Describe each of the fields of the TCP segment header format of Internet Transport Protocol in detail. **(5 Marks)**
6. Answer the following:
- (a) Explain Hierarchical Routing using any suitable network example. Give advantages and disadvantages of this routing scheme? **(7 Marks)**
 - (b) We use CIDR. Without using longest prefix matching, a forwarding table looks like this. If we use longest prefix matching, we can combine a few entries together. What is a table with a minimum number of entries that still be able to forward packet correctly? **(3 Marks)**

Prefix	Outgoing Interface
128.0.0.0/11	eth1
128.16.0.0/12	eth1
128.24.0.0/12	eth2
128.32.0.0/12	eth2
128.40.0.0/12	eth1
128.48.0.0/11	eth1
128.64.0.0/9	eth0
128.128.0.0/10	eth0
128.160.0.0/11	eth1
128.176.0.0/11	eth0
128.192.0.0/9	eth0
default	eth3